

Mini Project Report

Project Title: Diabetes Prediction Using Machine Learning

Team Member: 23051715, 23051712, 23051529, 23051536, 23052348

Domain: Machine Learning

Tools Used: Python, Pandas, NumPy, Scikit-learn, Matplotlib, seaborn

1. Introduction

This project focuses on predicting diabetes using Machine Learning models. Multiple classification algorithms were implemented and compared to select the most accurate and efficient model.

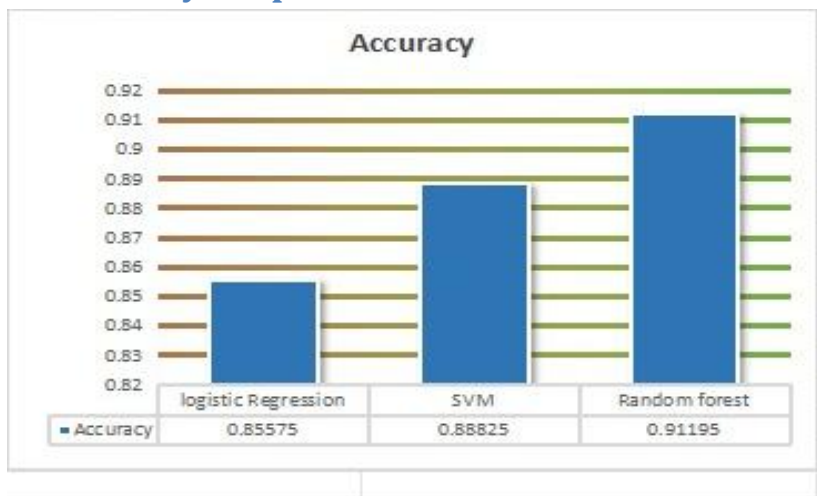
2. Models Implemented

- Logistic Regression
- Support Vector Machine (SVM)
- Random Forest Classifier

3. Model Accuracy Comparison

Model	Accuracy
Logistic Regression	0.85575
SVM	0.88825
Random Forest	0.91195

4. Accuracy Graph



5. Logistic Regression Implementation

```
from sklearn.linear_model import LogisticRegression
lr = LogisticRegression()
lr.fit(x_train,y_train)
```

[15]

...

LogisticRegression		
▼ Parameters		
penalty		'l2'
dual		False
tol		0.0001
C		1.0
fit_intercept		True
intercept_scaling		1
class_weight		None
random_state		None
solver		'lbfgs'
max_iter		100

6. Random Forest Implementation

```
rf = RandomForestClassifier(n_estimators = 100,random_state=42)
rf.fit(X_train,y_train)
```

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...

▼ RandomForestClassifier

► Parameters

```
pred2 = rf.predict(X_test)
pred2
```

[30]

array([0, 1, 0, ..., 0, 0, 0], shape=(20001,))

```
print('Accuracy:',accuracy_score(y_test,pred2))
print('Recall:',recall_score(y_test,pred2))
print('Precision:',precision_score(y_test,pred2))
print('F1 Score:',f1_score(y_test,pred2))
```

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...

Accuracy: 0.911954402279886
Recall: 0.7514450867052023
Precision: 0.8106001558846454
F1 Score: 0.7799025121859767

7. SVM Implementation

```
> from sklearn.metrics import accuracy_score, recall_score, precision_score, f1_score

acc = accuracy_score(pred,y_test)
re = recall_score(pred,y_test)
pr = precision_score(pred,y_test)
f1 = f1_score(pred,y_test)

print('Accuracy:',acc)
print('Recall:',re)
print('Precision:',pr)
print('F1-Score:',f1)

[18]
... Accuracy: 0.855757212139393
Recall: 0.7161203109158499
Precision: 0.5088856868395774
F1-Score: 0.5949740277972765
```

8. Conclusion

In this study, we systematically evaluated three machine learning models—Random Forest, Logistic Regression, and Support Vector Machine (SVM)—for diabetes prediction. To ensure robustness and generalizability, the models were tested on *two different lifestyle-based diabetes datasets sourced from Kaggle*, each containing unique distributions and feature sets. This dual-dataset approach allowed us to assess the adaptability of the models across varying data characteristics.

Across both datasets, *Random Forest consistently achieved the highest accuracy and superior performance metrics*, including precision, recall, and F1-score. Its ensemble learning mechanism enabled it to effectively capture complex, non-linear relationships between lifestyle factors and diabetes outcomes. In contrast, Logistic Regression and SVM, while yielding reasonable results, struggled to maintain consistent performance across the datasets, highlighting their limitations in handling diverse feature interactions.

Based on these findings, *Random Forest was selected as the final model for diabetes prediction*. Its ability to deliver reliable results across multiple datasets underscores its robustness and suitability for healthcare applications, where predictive accuracy and consistency across diverse patient populations are critical. This conclusion not only validates Random Forest as the optimal choice for this task but also emphasizes the importance of evaluating models on multiple datasets to ensure generalizability and real-world applicability.